

UPI Transaction Analysis

Vinay Pratap Vishwakarma
Data Analyst



Overview

The UPI Transaction Analysis project focuses on analyzing digital payment data to identify fraud risks, transaction failures, and customer behaviour patterns. Using SQL, Python, and Power BI, the project performs data cleaning, exploratory analysis, and KPI development to generate actionable insights.

Interactive dashboards were built to monitor key metrics such as transaction volume, fraud rate, and failure rate, helping stakeholders make data-driven decisions and improve platform performance.

Problem Statement

The rapid growth of UPI transactions has increased challenges related to transaction failures, fraud detection, and customer churn. Issues such as bank downtime, network errors, and security risks impact customer experience and operational efficiency. This project analyses UPI transaction data using Power BI dashboards to identify trends, detect fraud patterns, reduce failures, and improve customer retention through data-driven insights.

Tools & Methodology

A structured end-to-end pipeline transforms raw transaction data into interactive, decision-ready dashboards.

Excel : **Exploratory summaries, scenario modeling**

Python : **Data wrangling, EDA & Hypothesis Testing**

SQL : **Data extraction, joins, aggregations from relational databases**

Power BI : **Dashboard with KPIs, visual analytics, and filters**



Excel Overview & Task Workflow

Data Quality



Validation & Quality Checks

- Verified foreign key relationships across customer, transaction, merchant, device, UPI account, and fraud datasets.
- Identified missing or null values in critical business fields such as:



Transaction Amount



Status



Fraud Flag



Risk Score



Device Type

- Detected and corrected inconsistent categorical values and data entry errors.



Data Quality Monitoring

- Used Excel Conditional Formatting to highlight anomalies, missing values, and invalid entries.
- Developed a centralized Data Quality Log capturing:



Affected columns



Type of inconsistency



Recommended corrective actions



Count & % of impacted records

- Standardized date and categorical formats across all tables for consistency.
- Implemented a Validation Status flag to identify records requiring further review or remediation.



OUTCOME

Enhanced overall data reliability, improved consistency across relational datasets, and established a strong foundation for accurate reporting, fraud detection, transaction analysis, and customer behaviour insights.



SQL Overview & Task Workflows

1

CSV to SQL DDL



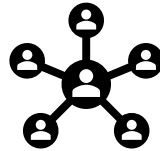
2

Create Table & Import Data



3

Data Modelling



4

Write Queries



5

Business Insights



6

Outcomes



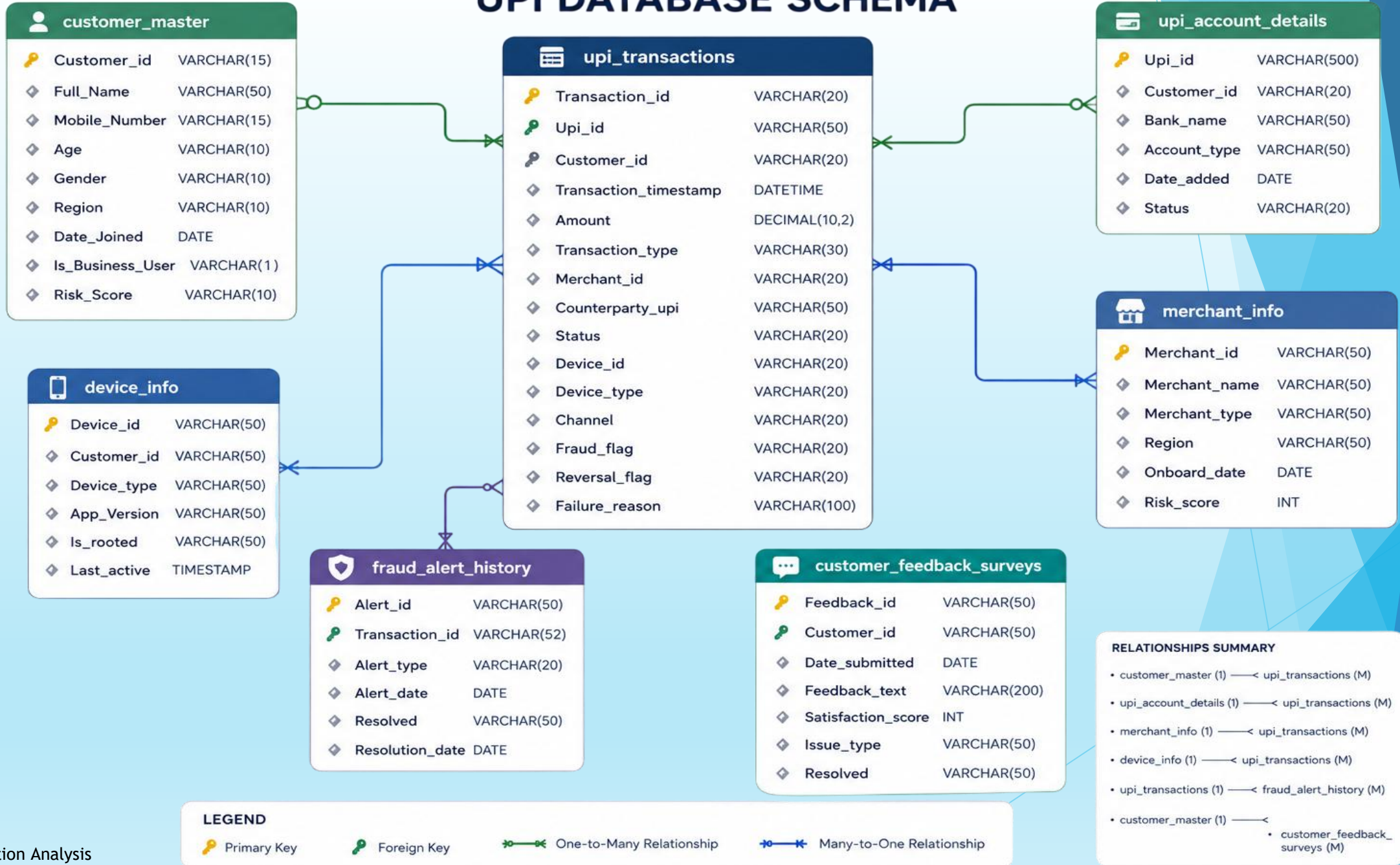
Top 5 Business Insights

1. UPI transactions showed rapid growth, reaching peak activity in 2025.
2. Transaction success rate remained strong at **92.1%**, though failures still impacted user experience.
3. Fraud cases increased significantly with transaction growth, especially in QR-based payments.
4. Customer churn reached **13.37%**, highlighting retention challenges.
5. North region and top banks like Kotak & SBI led transaction volume, while some regions showed higher fraud and failure rates.

Sample Query

```
1 SELECT
2     D.is_rooted,
3     COUNT(T.transaction_id) AS Count_of_Failed_Transactions
4 FROM
5     device_info D
6 JOIN
7     upi_transaction_history T ON D.device_id = T.device_id
8 WHERE
9     T.status = 'Failed'
10 GROUP BY D.is_rooted
11 ORDER BY COUNT(T.transaction_id) DESC;
```


UPI DATABASE SCHEMA



Most Important Queries

Account type vs fraud

```
1 SELECT
2     A.account_type,
3     SUM(T.fraud_flag) AS Count_Of_Fraud_Transactions
4 FROM
5     upi_account_details AS A
6     RIGHT JOIN
7     upi_transactions AS T ON T.upi_id = A.upi_id
8 GROUP BY A.account_type
9 ORDER BY SUM(T.fraud_flag) DESC;
```

account_type	Count_Of_Fraud_Transactions
savings	685
current	676
credit_card_linked	639

Bank vs Fraud

```
3 SELECT
4     A.bank_name,
5     SUM(T.Fraud_Flag) AS Count_Of_Fraud_Transactions
6 FROM
7     upi_account_details A
8     JOIN
9     upi_transactions T ON T.upi_id = A.upi_id
10 GROUP BY A.bank_name
11 ORDER BY SUM(T.Fraud_Flag) DESC;
```

bank_name	Count_Of_Fraud_Transactions
pnb	374
kotak	354
hdfc	334
sbi	331
axis	310
icici	297

Gender vs Transaction

```
34 SELECT
35     C.gender, COUNT(T.Fraud_flag) AS Count_Of_Fraud_Transactions
36 FROM
37     upi_transactions T
38     JOIN
39     customer_master C ON C.customer_id = T.customer_id
40 WHERE
41     Fraud_flag = 1
42 GROUP BY C.gender;
```

gender	Count_Of_Fraud_Transactions
female	974
male	987
other	39

Channel vs fraud

```
1 SELECT
2     T.channel, COUNT(F.alert_id) AS Count_Of_Fraud_Transactions
3 FROM
4     fraud_alert_history F
5     JOIN
6     upi_transactions T ON T.transaction_id = F.transaction_id
7 GROUP BY t.channel
8 ORDER BY COUNT(F.alert_id) DESC;
```

channel	Count_Of_Fraud_Transactions
qr_code	698
app	673
intent	629

Device vs fraud

```
34 SELECT
35     T.device_type,
36     COUNT(F.alert_id) AS Count_Of_Fraud_Transactions
37 FROM
38     fraud_alert_history F
39     JOIN
40     upi_transactions T ON T.transaction_id = F.transaction_id
41 GROUP BY t.device_type
42 ORDER BY COUNT(F.alert_id) DESC;
```

device_type	Count_Of_Fraud_Transactions
feature_phone	551
android	501
tablet	488
ios	460

Account vs Transactions

```
33 SELECT
34     A.account_type,
35     COUNT(T.transaction_id) AS Count_Of_Failed_Transactions
36 FROM
37     upi_account_details A
38     JOIN
39     upi_transactions T ON A.upi_id = T.upi_id
40 WHERE
41     T.status = 'Failed'
42 GROUP BY A.account_type
43 ORDER BY COUNT(T.transaction_id) DESC;
```

account_type	Count_Of_Failed_Transactions
current	2006
savings	1952
credit_card_linked	1913

Python Overview & Task Workflows

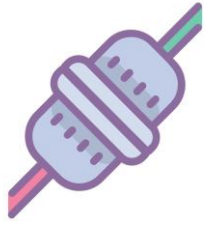
1

Import Library



2

Create Engine



3

Connect With
SQL Database



4

Write Python
Script



5

Business
Insights



6

Outcomes



Business Insights

1. Transaction behavior varies significantly across different device types.
2. Fraud occurrence differs by region, indicating regional risk variation.
3. Certain device types are more associated with fraudulent transactions.
4. Customer risk score shows very weak correlation with actual fraud activity.
5. Integrated datasets enabled deeper analytical and statistical insights.

Business Recommendation

1. Implement device-specific monitoring and user experience optimization.
2. Strengthen fraud controls in high-risk regions.
3. Apply stricter security measures for high-risk device categories.
4. Improve the customer risk scoring model using behavioural data.
5. Develop predictive analytics models for fraud and risk detection.

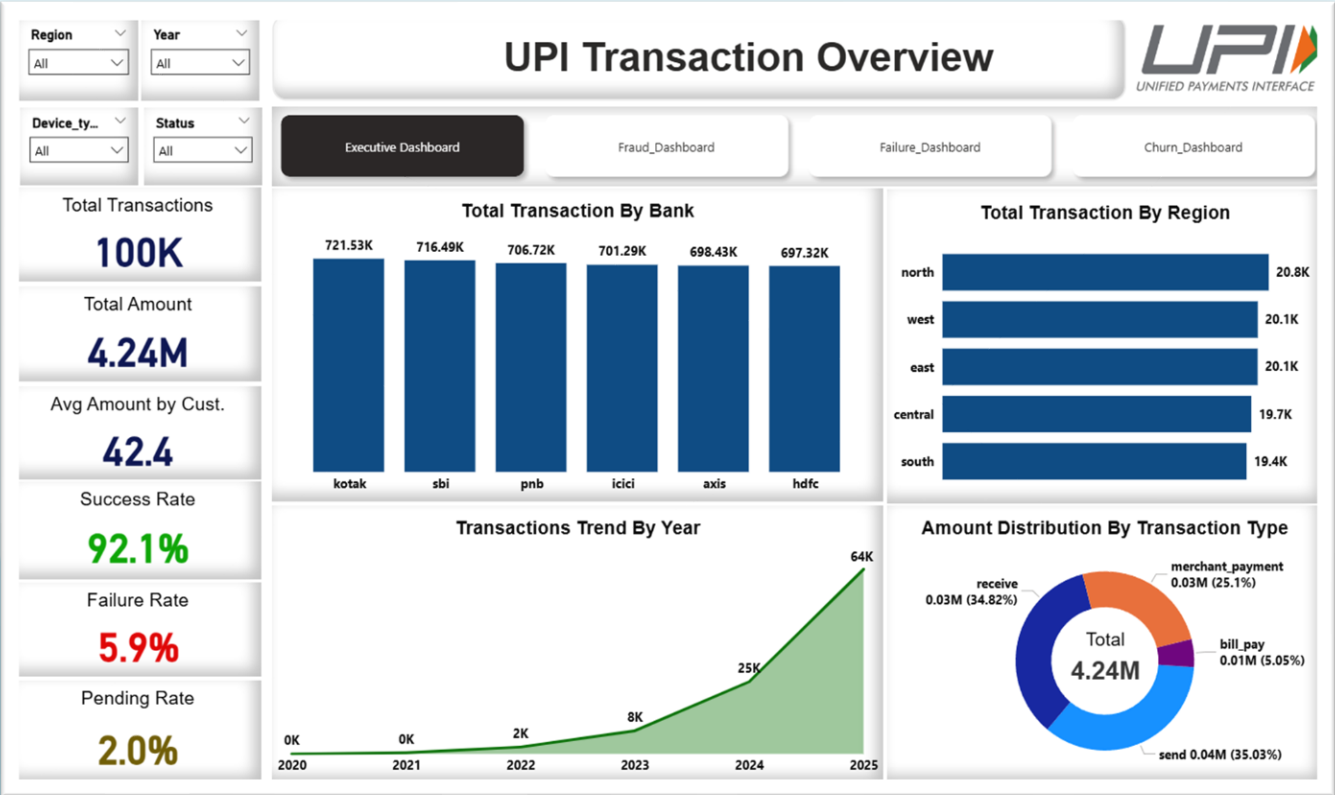
Executive Summary & Recommendations

Business Insights

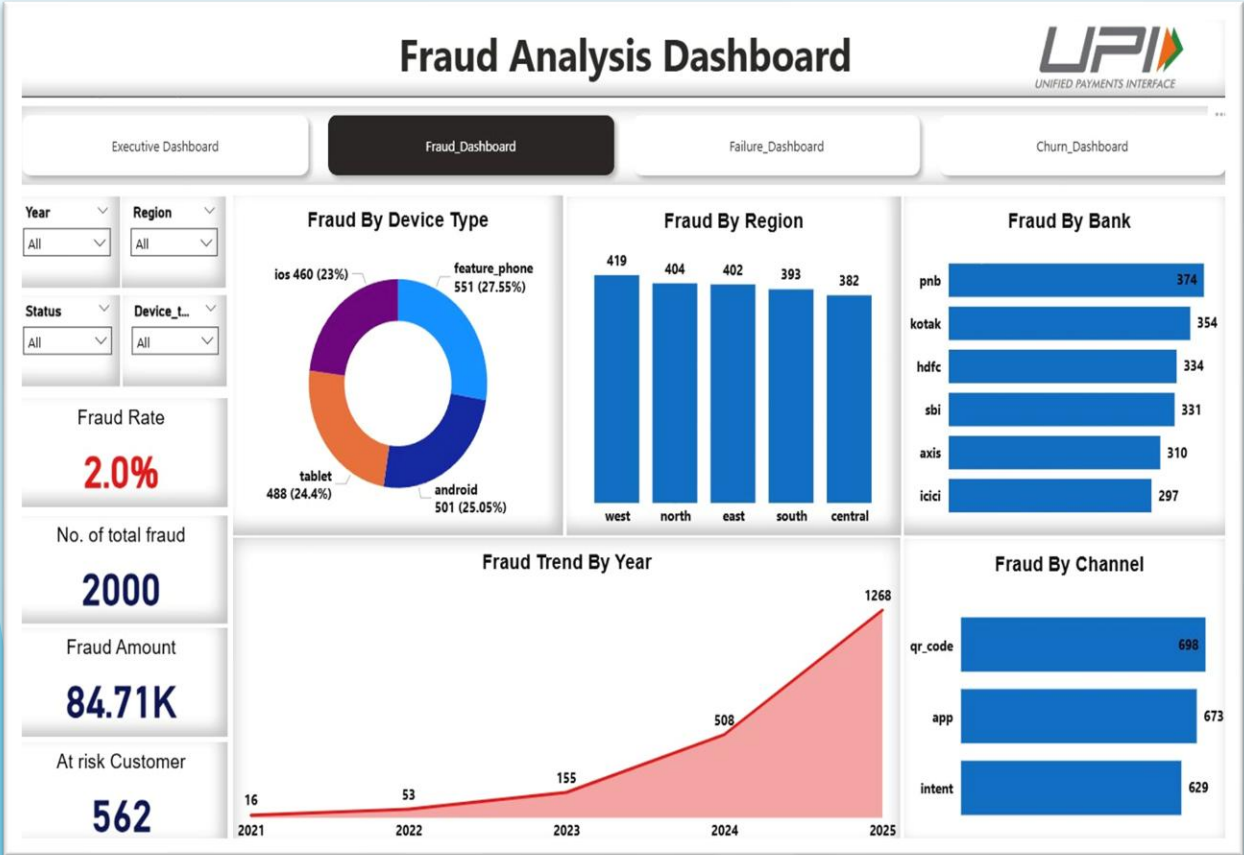
- 1. Total transactions reached 100K with a transaction value of 4.24M. 92.1% success rate indicates strong platform reliability.
- 2. Kotak Bank recorded the highest transaction volume among banks. North region leads in transaction activity, while South shows lower engagement.
- 3. Transactions grew rapidly after 2023, reaching 64K in 2025. Send & Receive transactions dominate the ecosystem (~70% share).
- 4. Bill payments contribute the least, indicating low adoption.

Recommendation

- 1. Reduce failure rate (5.9%) through bank/API performance optimization.
- 2. Promote bill payments & merchant transactions using cashback and rewards.
- 3. Focus growth campaigns in low-performing regions.
- 4. Strengthen fraud monitoring systems as transaction volume scales rapidly.



Fraud Summary & Recommendations



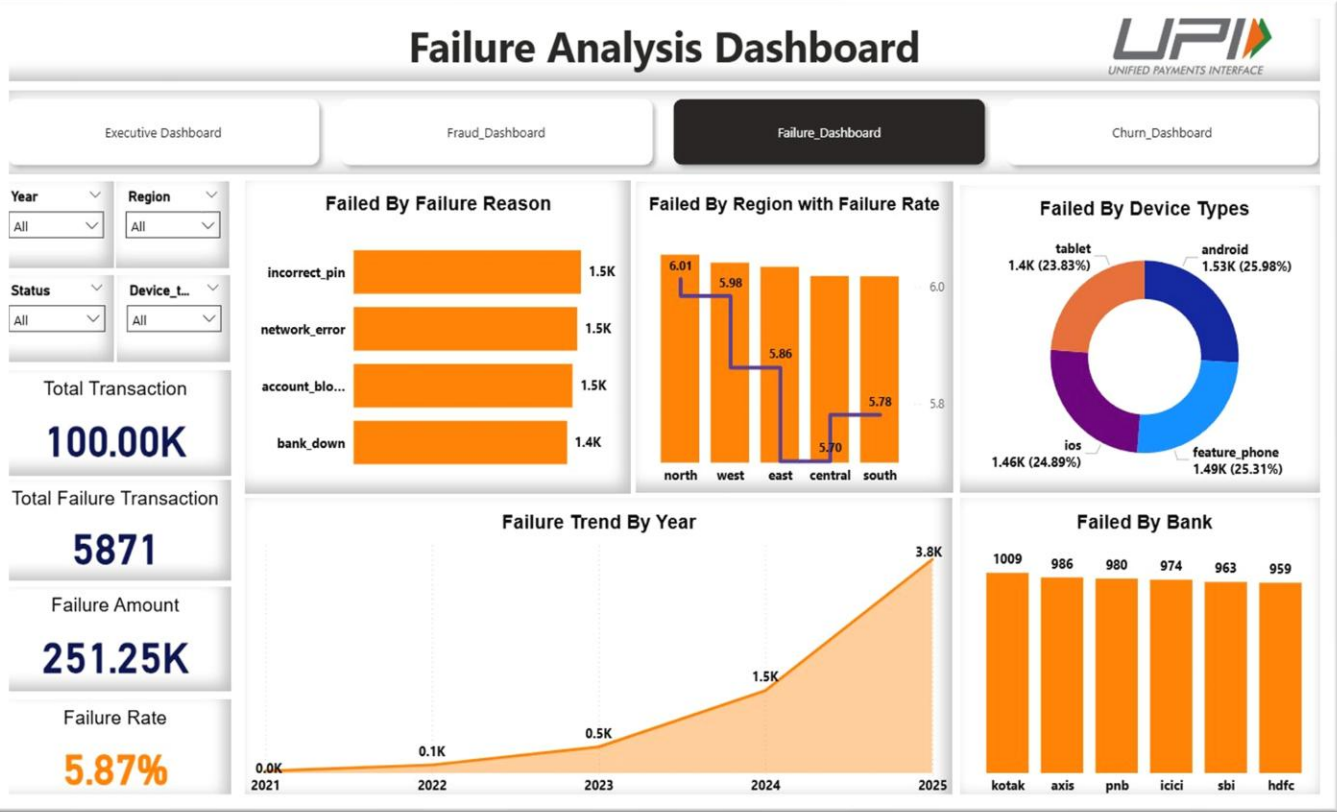
Business Insights

1. Overall fraud rate stands at 2.0% with 2,000 fraud cases detected.
2. Total fraud amount reached 84.71K, impacting 562 at-risk customers.
3. Fraud cases increased sharply from 16 (2021) to 1,268 (2025), indicating rising security threats.
4. PNB recorded the highest fraud activity, followed by Kotak and HDFC.
5. West region shows the highest fraud volume, while Central region has the lowest.
6. Fraud is widely distributed across all device types, with feature phones contributing the highest share.
7. QR Code channel experienced the highest fraud activity compared to App and Intent channels.

Recommendation

1. Strengthen real-time fraud detection and anomaly monitoring systems.
2. Increase security checks for QR Code transactions and high-risk banks. Implement device-based authentication for feature phone and Android users.
3. Conduct region-specific fraud awareness campaigns in high-risk areas.
4. Use AI/ML models for early fraud prediction and suspicious transaction alerts.

Failure Summary & Recommendations



Business Insights

1. Total failed transactions reached 5,871 with a failure rate of 5.87%.
2. Failure amount impacted transactions worth 251.25K.
3. Transaction failures increased significantly, reaching 3.8K in 2025.
4. North region recorded the highest failure rate (6.01%). Android and Feature Phone users show slightly higher failure contribution.
5. Kotak Bank experienced the highest number of failed transactions.

Recommendation

1. Improve bank server uptime and network reliability to reduce transaction failures.
2. Introduce smart alerts and retry mechanisms for failed payments.
3. Strengthen customer education for PIN-related transaction errors.
4. Monitor high-failure regions and banks with real-time operational dashboards.
5. Optimize mobile app performance across Android and feature phone devices.

Churn Summary & Recommendations

Business Insights

1. Total customer base reached 10K, with 1,337 churned customers and a churn rate of 13.37%.
2. Active customers increased significantly over the years, showing strong platform growth.
3. Central region recorded the highest churn, while South had the lowest churn count.
4. iOS users showed the highest churn among device types.
5. Churn distribution is almost equal between male (48.84%) and female (49.07%) customers.
6. Churn trend increased gradually alongside customer growth, indicating retention challenges at scale.

Recommendation

1. Launch targeted retention campaigns for high-churn regions and device segments.
2. Improve customer experience by reducing transaction failures and fraud-related issues.
3. Introduce loyalty rewards, cashback, and engagement programs for inactive users.
4. Use predictive churn models to identify at-risk customers early.
5. Monitor customer activity trends and personalize offers to improve retention.

Churn Analysis Dashboard



Executive Dashboard

Fraud_Dashboard

Failure_Dashboard

Churn_Dashboard

Year: All
Region: All
Status: All
Device_t...: All

Total Customer

10.00K

Active Customer

5695

Churned Customer

1337

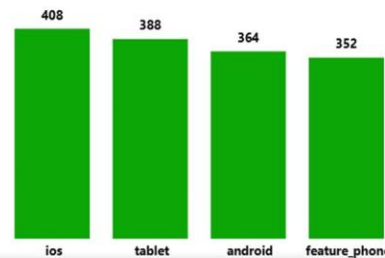
Churned Rate

13.37%

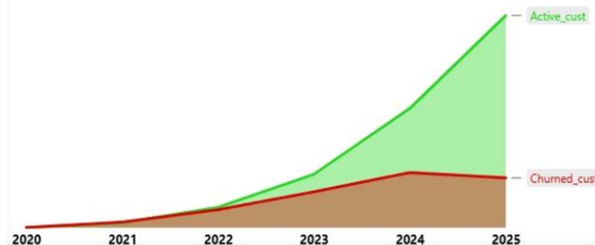
Churn By Reason



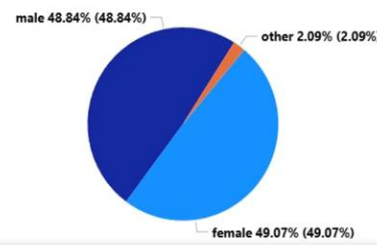
Churn By Device Types



Active Vs Churned Customers Trend by Year



Churn By Gender



Thank you

Vinay Pratap Vishwakarma

pratapvinay365@gmail.com

<https://github.com/VinayPratap18>

[LinkedIn/vinay-pratap-vishwakarma](#)